

INTERNSHIP REPORT

ON

FIBER OPTIC COMMUNICATIONS, FTTH NETWORK ARCHITECTURE AND ELECTRICAL HOME APPLIANCES MAINTENANCE

Submitted by:
Suhana.K

In partial fulfillment of the requirements for the award of the Degree of
Bachelor of Science in Physics

Under the Faculty Guidance of:
Dr. Mohemmed Shanid N A
Department of Physics



Department of Physics
Sullamussalam Science College, Areekode
(Affiliated to Calicut University)

In Industrial Association With:
Bfone Fibernet, Edavannappara

Date of Submission: June 14, 2026

DEPARTMENT OF PHYSICS
SULLAMUSSALAM SCIENCE COLLEGE, AREEKODE

Areekode, Malappuram, Kerala - 673639

CERTIFICATE

This is to certify that the internship report entitled "**Fiber Optic Communications and FTTH Network Architecture**" is a bona fide record of the industrial training carried out by **Suhana.K** under my academic supervision and guidance, in corporate collaboration with **Bfone Fibernet, Edavannappara**, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Science in Physics.

Dr. Mohemmed Shanid N A
Internal Faculty Guide and Department of
Physics

Head of the Department

and

External Examiner

Internal Examiner

Contents

I	Fiber Optic Communications and FTTH Network Architecture	6
	Part Introduction: Fiber Optic Infrastructure	7
1	Introduction to Optical Fiber Communication	8
1.1	Overview	8
1.2	The Basic Fiber Optic Link	8
1.3	Advantages of Fiber Optics	8
2	Principle and Cable Structure	9
2.1	Principle of Operation: Total Internal Reflection	9
2.2	Cable Anatomy	9
2.3	Cable Types	9
3	Link Budgeting, Splicing, and Mathematical Calculations	10
3.1	OFC Splicing	10
3.2	Logarithmic Conversion Mechanics	10
3.3	Splitter Loss Calculations	10
3.4	Standard Loss Values for Link Performance	10
3.5	Mathematical Verification: Power Link Budget Case Study	10
3.5.1	Case Study Parameter Blueprint	11
3.5.2	Link Status Analysis	11
4	Optical Fiber Splicing: Detailed Analysis	12
4.1	What is Splicing?	12
4.2	Types of Splicing	12
4.2.1	Fusion Splicing	12
4.2.2	Mechanical Splicing	12
4.3	Importance and Significance	12
4.4	Step-by-Step Fusion Splicing Process	13
4.5	Advantages	13
4.6	Future Scope	13
4.7	Conclusion	13

5	Introduction to FTTH & Passive Optical Networks (PON)	14
5.1	Overview of FTTH	14
5.2	The FTTH Network Architecture Layers	14
5.2.1	A. The Feeder Segment	14
5.2.2	B. The Distribution Segment	14
5.2.3	C. The Drop Segment	15
5.3	Core Hardware Components	15
5.3.1	Active Components (Power Required)	15
5.3.2	Passive Components (No Power Required)	15
6	FTTH Deployment, Coding and Testing	16
6.1	Fiber Color Coding Standard (TIA-598-C)	16
6.2	Physical Deployment & Site Survey	16
6.2.1	Site Survey & Planning	16
6.2.2	Outdoor Routing Operations	17
6.2.3	Indoor Termination	17
6.3	Splicing Methods Comparison	17
6.4	Link Budget & Testing Benchmarks	17
6.4.1	Wavelength Multiplexing	17
6.4.2	Testing Parameters	17
6.5	Troubleshooting & Indicators	18
7	Field Visit: BSNL Exchange Office, Manjeri	19
7.1	Introduction to the Field Visit	19
7.2	Observations and Core Infrastructure	19
7.3	Conclusion of the Visit	19
	Part Conclusion: Fiber Optic Infrastructure	20
II	Electrical Home Appliances Maintenance and Troubleshooting	21
	Part Introduction: Domestic Electrical Engineering	22
8	Fundamentals of Electrical Systems & Safety	23
8.1	Introduction	23
8.2	Fundamentals of Electrical Systems & Safety Overview	23
8.2.1	Circuit Behaviors & Laws	23
8.2.2	Earthing and Protection Systems	24
9	Technical Breakdown & Diagnostics of Core Home Appliances	25
9.1	Electric Iron	25
9.2	Induction Cooker	26
9.3	Mixer Grinder	26
9.3.1	Ceiling Fan	27

9.3.2 Wall Fan	27
10 Troubleshooting & Maintenance Practice Matrix	29
10.1 Internship Outcomes & Skills Acquired	29
11 Applied Practical Calculations	31
11.1 Domestic Electricity Consumption Auditing	31
11.2 Charge Calculation in Audio Systems	31
Part Conclusion: Domestic Electrical Engineering	32

INTERNSHIP ACTIVITY SCHEDULE

Table 1: Day-wise Internship Activity Log

Day	Date	Topics / Activities Covered	Remarks
Day 1	21.05.2026	Optical Fiber Theory Class (Introduction to OFC)	
Day 2	22.05.2026	Splicing Practical Section	
Day 3	23.05.2026	Repairing of the Ceiling Fan	
Day 4	25.05.2026	• Theory Class of Fiber To The Home (FTTH) • Repairing of Mixer Grinder • Field Visit of Splicing	
Day 5	26.05.2026	Field Visit at Manjeri Telephone Exchange	
Day 6	08.06.2026	Power Calculation Theory Class	
Day 7	09.06.2026	• Repairing of Iron Box • Repairing of Wall Fan • Repairing of Induction Cooker	

Part I

Fiber Optic Communications and FTTH Network Architecture

Part Introduction: Fiber Optic Infrastructure

The modern telecommunication landscape is fundamentally powered by high-speed data transmission systems that rely on optical physics. Transitioning away from traditional legacy copper wire infrastructures, optical fibers utilize light pulses to transmit vast amounts of data over immense distances with minimal attenuation. This part of the internship report establishes a detailed technical study of modern Fiber-Optic Communication (OFC) and Fiber-to-the-Home (FTTH) network architectures, directly informed by industrial training with Bfone Fibernet.

The primary objective of this module is to comprehensively trace the optical signal pathway from its structural origins to real-world residential deployment. The chapters herein analyze the physical principles of Total Internal Reflection (TIR) that guide light within glass matrices, the structural mechanics of modern multi-core cables, and the quantitative engineering required for link budgeting and structural calculations. Furthermore, this section details practical operational protocols, including high-precision fusion splicing, color-coding standards, network diagnostics, and troubleshooting methodologies implemented in active field environments. This comprehensive study concludes with a field analysis of a regional telecommunication distribution center at the BSNL Exchange Office.

Chapter 1

Introduction to Optical Fiber Communication

1.1 Overview

Optical Fiber Communication (OFC) is a method of transmitting information from one place to another by sending pulses of light through an optical fiber. The light forms an electromagnetic carrier wave that is modulated to carry information.

1.2 The Basic Fiber Optic Link

A typical optical fiber communication link consists of several key components:

- **Encoder/Source:** Converts data into electrical signals.
- **Transmitter (E/O):** Converts electrical signals into optical signals using LEDs or LASERs.
- **Optical Fiber:** The medium that guides the light signal.
- **Receiver (O/E):** Uses a photo-detector (PIN or APD) to convert light back into electrical signals.
- **Decoder:** Recovers the original data from the signal.

1.3 Advantages of Fiber Optics

- **Non-conductive:** No risk of sparks or electrical hazards.
- **Electromagnetic Immunity:** Immune to EMI and lightning.
- **Large Bandwidth:** Can carry massive amounts of data.
- **Low Loss:** Signal attenuation is significantly lower than copper cables.
- **Small and Lightweight:** Easier to install and transport.

Chapter 2

Principle and Cable Structure

2.1 Principle of Operation: Total Internal Reflection

Light travels through the fiber core based on the principle of **Total Internal Reflection (TIR)**. For TIR to occur:

1. The refractive index of the core (n_1) must be greater than the cladding (n_2).
2. The angle of incidence must be greater than the critical angle (θ_c).

The critical angle is calculated as:

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) \quad (2.1)$$

2.2 Cable Anatomy

A standard OFC cable consists of:

- **Core:** The innermost glass or plastic part where light travels.
- **Cladding:** Layer with a lower refractive index that reflects light back into the core.
- **Buffer:** Protects the fiber from moisture.
- **Strength Member:** Usually Aramid yarn (Kevlar) to prevent cable breakage.
- **Outer Jacket:** Protects against environmental damage.

2.3 Cable Types

- **Tight-Buffer Cable:** Used mainly for indoor applications and patch cords.
- **Loose-Tube Cable:** Used for outdoor environments (ducts, direct burial).
- **Ribbon Cable:** Dozens of fibers packed together for high-density applications.

Chapter 3

Link Budgeting, Splicing, and Mathematical Calculations

3.1 OFC Splicing

Splices are permanent connections between two fibers. The main methods are:

- **Fusion Splicing:** Using an electric arc to melt and fuse fibers.
- **Mechanical Splicing:** Aligning fibers using a mechanical fixture.

3.2 Logarithmic Conversion Mechanics

Optical power values are measured using the logarithmic decibel-milliwatt (dBm) scale to make it easy to calculate losses across long fiber links:

$$P_{\text{dBm}} = 10 \log_{10} \left(\frac{P_{\text{mW}}}{1 \text{ mW}} \right) \quad (3.1)$$

3.3 Splitter Loss Calculations

The theoretical insertion loss of an optical splitter scales based on the number of output ports (N):

$$\text{Insertion Loss}_{\text{Ideal}} = 10 \log_{10}(N) \quad (3.2)$$

3.4 Standard Loss Values for Link Performance

To perform reliable link calculations, the engineering team uses the following standard attenuation parameters:

3.5 Mathematical Verification: Power Link Budget Case Study

To calculate the received power (P_r) at an endpoint, use the following power budget equation:

$$P_r = P_t - \text{Loss}_{\text{total}} \quad (3.3)$$

Table 3.1: Standard Loss Parameters Used in FTTH Networks

Network Element Component	Standard Loss Value
Fiber Path Loss @ 1310 nm / 1490 nm	0.40 dB per km
Fiber Path Loss @ 1550 nm	0.25 dB per km
Fusion Splice Junction	0.10 dB per splice
SC/UPC Physical Connector Pair	0.35 dB per connector
SC/APC Physical Connector Pair	0.25 dB per connector
1 × 2 Splitter Module	3.0 dB average insertion loss
1 × 16 Splitter Module	13.0 dB average insertion loss
1 × 32 Splitter Module	16.0 dB average insertion loss

3.5.1 Case Study Parameter Blueprint

Consider a regional fiber run managed by Bfone Fibernet spanning a total distance of 7 km. The link includes 4 fusion splices, 2 SC/UPC connector pairs, a 1 × 16 splitter, and a 1 × 8 cascading splitter module.

The total link budget metrics break down into the following distinct attenuation values:

- **Fiber Line Attenuation Loss:** $7 \text{ km} \times 0.40 \text{ dB/km} = 2.8 \text{ dB}$
- **First Stage Insertion Loss** (1 × 16 Splitter) = 13.0 dB
- **Second Stage Insertion Loss** (1 × 8 Splitter) = 10.0 dB
- **Total Mechanical Splice Losses:** $4 \times 0.10 \text{ dB} = 0.4 \text{ dB}$
- **Total Dynamic Connector Losses:** $2 \times 0.35 \text{ dB} = 0.7 \text{ dB}$

Summing these individual loss values yields the total attenuation for the link:

$$\begin{aligned} \text{Total Attenuation Loss} &= 2.8 + 13.0 + 10.0 + 0.4 + 0.7 \\ &= \mathbf{26.9 \text{ dB}} \end{aligned}$$

3.5.2 Link Status Analysis

Standard ONT modems require an input signal level within an optical power range between −10 dBm and −28 dBm to operate reliably:

$$\text{Receiver Sensitivity Threshold} \leq -28 \text{ dBm} \quad (3.4)$$

Because the calculated total loss for this link is 26.9 dB, a typical transmitter power output of 0 dBm ensures that the received power level stays above the receiver's minimum sensitivity threshold ($-26.9 \text{ dBm} \geq -28 \text{ dBm}$). This mathematical verification confirms that the fiber link will function correctly and maintain high signal reliability.

Chapter 4

Optical Fiber Splicing: Detailed Analysis

4.1 What is Splicing?

Splicing is the process of joining two optical fiber cores permanently to create a continuous optical path. Unlike connectors, which are designed for frequent re-mating, splices are meant to be enduring and stable joints. The primary goal is to minimize optical loss and prevent back-reflection.

4.2 Types of Splicing

4.2.1 Fusion Splicing

This is the most reliable and industry-standard method. It involves using an electric arc to melt and fuse two fiber ends together.

- **Loss:** Typically 0.01 to 0.05 dB.
- **Usage:** Backbone networks, long-haul links, and FTTH.

4.2.2 Mechanical Splicing

This method uses a mechanical fixture to align the fiber ends and an index-matching gel to facilitate light transmission between the cores.

- **Loss:** Typically 0.2 to 0.5 dB.
- **Usage:** Emergency repairs, temporary setups, and environments where fusion equipment is not available.

4.3 Importance and Significance

- **Link Extension:** Connecting multiple segments to build long-distance networks.
- **Signal Integrity:** Maintaining near-perfect light transmission with minimal attenuation.

- **Back-reflection Control:** Minimizing return loss, which protects laser sources.
- **Durability:** Providing maintenance-free connections that withstand environmental stress.

4.4 Step-by-Step Fusion Splicing Process

1. **Stripping:** The 250 μm outer coating is removed using specialized fiber strippers without scratching the glass.
2. **Cleaning:** The bare glass fiber is thoroughly cleaned using 99% Isopropyl Alcohol (IPA) and lint-free wipes to remove all contaminants.
3. **Cleaving:** The fiber is cut using a precision diamond-blade cleaver to ensure a perfect 90-degree end face (angle $< 0.5^\circ$).
4. **Fusion:** The fiber ends are aligned using the Profile Alignment System (PAS) and fused using an electric arc at 2000°C.
5. **Protection:** A heat-shrink splice sleeve is applied and heated to protect the joint from physical damage and moisture.

4.5 Advantages

- Lowest possible insertion loss.
- Permanent and highly reliable joints.
- Resistant to vibration and temperature fluctuations.
- Space-efficient for high-density fiber enclosures.

4.6 Future Scope

With the rapid evolution of optical networks, the scope of fiber splicing is expanding towards:

- **Automation:** Developing fully autonomous AI-driven splicing robots for sub-sea and hazardous environments.
- **Multi-core Fiber Splicing:** Moving beyond single-core to support high-density multi-core fibers to increase bandwidth capacity.
- **Enhanced Precision:** Integrating advanced nanotechnology for near-zero loss splicing in extreme environments.

4.7 Conclusion

Optical fiber splicing is the foundation of modern high-speed communication. By ensuring a seamless and permanent joint, it guarantees that data travels across vast distances with negligible loss. While fusion splicing is the preferred method for permanent infrastructure, mechanical splicing remains a vital tool for rapid, field-based repairs.

Chapter 5

Introduction to FTTH & Passive Optical Networks (PON)

5.1 Overview of FTTH

Fiber-to-the-Home (FTTH) represents the ultimate evolutionary step in broadband access networks, replacing traditional copper-based infrastructures (like DSL or Coaxial cables) entirely with optical fiber lines directly to individual subscriber premises.

Most modern FTTH networks utilize **PON (Passive Optical Network)** technology. The term “passive” is critical: it means the outer distribution network requires no electrical power between the provider’s central exchange and the customer’s home. Signals are routed and split using optical principles alone, drastically lowering operational costs, minimizing equipment failure due to power surges, and maximizing lifespan. This point-to-multipoint architecture allows a single optical fiber to serve multiple premises through splitters.

5.2 The FTTH Network Architecture Layers

The physical infrastructure of an FTTH network is divided into three distinct geographical segments:

5.2.1 A. The Feeder Segment

- **Origin & Termination:** Starts at the service provider’s Central Office (CO) and runs to a local neighborhood hub known as the **Fiber Distribution Hub (FDH)** or Primary Cross-Connect box.
- **Characteristics:** This segment uses high-core-count cables (often containing 48 to 144+ fibers) designed to carry aggregated data over long distances (up to 20 kilometers) without splitting.

5.2.2 B. The Distribution Segment

- **Origin & Termination:** Travels from the FDH to smaller street-level boxes called **Optical Distribution Points (ODP)** or Fiber Access Terminals (FAT).

- **Characteristics:** At the FDH, the feeder fibers are connected to optical splitters. The split fibers branch out into the neighborhood. These cables have a medium core count (typically 12 to 24 fibers) and run down streets, wrapping around utility poles or inside underground ducts.

5.2.3 C. The Drop Segment

- **Origin & Termination:** The final link that bridges the ODP/FAT on the street directly into the customer's living room or office.
- **Characteristics:** This utilizes a **Drop Cable**, which is a highly specialized, lightweight, ruggedized fiber cable (usually containing only 1 or 2 fibers) engineered to survive tight bends and physical pulling.

5.3 Core Hardware Components

An FTTH network relies on a strict handshake between active hardware (requiring power) and passive hardware (no power).

5.3.1 Active Components (Power Required)

- **OLT (Optical Line Terminal):**
 - *Location:* Installed at the Central Office (CO).
 - *Function:* The central “brain” of the network. It receives traditional digital data (IP traffic, television signals, voice) from the core internet backbone and converts it into laser light pulses. It manages traffic scheduling, authentication, and bandwidth allocation for thousands of subscribers.
- **ONT / ONU (Optical Network Terminal / Unit):**
 - *Location:* Installed inside the subscriber's home.
 - *Function:* The translator. It receives the optical light signals from the drop cable and converts them back into standard electrical signals (Ethernet/Wi-Fi) for consumer devices.

5.3.2 Passive Components (No Power Required)

- **Optical Splitters:** The heart of PON. A passive glass prism component that takes a single incoming beam of light and splits it into multiple identical streams to serve multiple users. Standard split ratios are **1:8, 1:16, 1:32, or 1:64**.
- **ODF / FDH Hubs:** Environmental enclosures or cabinets that house splices and splitters, protecting connections from water, dust, and temperature changes.

Chapter 6

FTTH Deployment, Coding and Testing

6.1 Fiber Color Coding Standard (TIA-598-C)

To safely identify individual fibers inside a multi-core cable, technicians use the industry-standard 12-color coding sequence. For cables with more than 12 fibers, the sequence repeats with a physical black tracer/stripe.

Table 6.1: TIA-598-C Color Coding Identification

No.	Fiber Color	Symbol	No.	Fiber Color	Symbol
1	 Blue	BL	7	 Red	RD
2	 Orange	OR	8	 Black	BK
3	 Green	GR	9	 Yellow	YL
4	 Brown	BR	10	 Violet	VI
5	 Slate (Grey)	SL	11	 Rose (Pink)	RS
6	White	WH	12	 Aqua	AQ

6.2 Physical Deployment & Site Survey

6.2.1 Site Survey & Planning

Before installation, technicians perform a survey consisting of:

1. **Route Mapping:** Finding the shortest, most efficient path from street FAT to house.
2. **Structural Assessment:** Identifying building entry points and optimal indoor ONT placement.
3. **Safety Check:** Keeping clear of electrical mainlines to avoid physical interference.
4. **BOM Calculation:** Generating the exact Bill of Materials (cables, clamps, rosettes).

6.2.2 Outdoor Routing Operations

- **Aerial Deployment:** Fiber cables are strung between utility poles using tension/ anchor clamps. Slack loops must be left at ends to accommodate weather-induced physical shifts.
- **Underground Deployment:** Fiber is pulled or blown through buried HDPE (High-Density Polyethylene) conduits for maximum protection against environmental hazards.

6.2.3 Indoor Termination

The drop cable passes into the home and terminates inside an **Access Terminal Box (ATB)** or Fiber Rosette on the wall. A highly flexible, bend-insensitive patch cord (typically G.657 standard) connects the ATB to the ONT, preventing signal loss around tight corners.

6.3 Splicing Methods Comparison

Connecting optical fibers requires microscopic precision. Two methods are utilized:

Table 6.2: Comparison of Fusion and Mechanical Splicing

Feature	Fusion Splicing	Mechanical / Fast Connectors
Mechanism	Uses a high-voltage electric arc to melt and fuse glass tips permanently.	Physically aligns two fiber ends inside a sleeve using index-matching gel.
Signal Loss	Extremely low (< 0.05 dB to 0.1 dB).	Slightly higher (0.2 dB to 0.5 dB).
Pros & Cons	Most reliable, but requires an expensive fusion splicer machine.	Fast setup, low initial tool cost. Ideal for home drop termination.

6.4 Link Budget & Testing Benchmarks

6.4.1 Wavelength Multiplexing

FTTH operates over single-mode fiber using separate bidirectional wavelengths:

- **1490 nm:** Downstream data/ voice traffic (OLT to ONT).
- **1310 nm:** Upstream data upload traffic (ONT to OLT).
- **1550 nm:** Optional downstream video/TV overlay.

6.4.2 Testing Parameters

- **Optical Power Meter (OPM):** Measures absolute light strength at the subscriber end. Acceptable operational range is between **-15 dBm and -25 dBm**. Readings below **-27 dBm** constitute an under-power fault.
- **OTDR:** Measures backscattered light to map the entire physical link and locate precise coordinates of cuts or defects.

6.5 Troubleshooting & Indicators

Field diagnostics rely heavily on the ONT hardware status LED combinations:

Table 6.3: ONT Troubleshooting Matrix

ONT Indicator	status State	Meaning / Corrective Action
LOS (Red)	Steady Glow	Loss of Signal: Total physical fiber cut or severe macro-bend. Check outside plant with OTDR.
PON (Green)	Blinking / Flashing	Registering: Authentication in progress. If persistent, light power is weak or configuration profile is missing.
PON (Green)	Steady Solid Glow	Connected: Symmetrical operational link established successfully.
Power Levels	Below -27 dBm	Power Fault: Dirty connector faces or high-loss splice. Rework connection or clean with a One-Click cleaner.

Chapter 7

Field Visit: BSNL Exchange Office, Manjeri

7.1 Introduction to the Field Visit

As part of the internship in optical fiber communication, a technical visit was conducted to the BSNL Exchange office, Manjeri to gain practical exposure to the operation and management of fiber-optic communication networks. The visit provided an opportunity to observe how telecommunication services are delivered from the exchange to end users through optical fiber infrastructure.

7.2 Observations and Core Infrastructure

During the visit, we observed different equipment used in the exchange, including the Optical Line Terminal (OLT), fiber distribution panels, and other networking devices. The officials explained how internet, voice, and other communication services are transmitted through optical fiber cables from the exchange to customers.

Behind the OLT, several backend devices are used for network management. The BNG (Broadband Network Gateway) handles user authentication, IP address allocation, routing, and internet session management. The MNGN (Multiplay Converged packet access Networks) is a secure, carrier-grade packet transmission network deployed within the local exchange. It supports services such as FTTH broadband, leased lines, and enterprise connections over a common network infrastructure. The OLT delivers internet services to ONTs through optical fiber cables.

7.3 Conclusion of the Visit

The visit provided practical knowledge about the working of a telecommunication exchange and helped us relate the theoretical concepts of optical fiber communication to real-world applications. It was a valuable learning experience that improved our understanding of modern fiber-optic communication systems.

Part Conclusion: Fiber Optic Infrastructure

In conclusion, this module has successfully detailed the comprehensive operational framework of modern high-speed telecommunications. Beginning with the core waveguiding physics of single-mode fibers, the report systematically evaluated how light pulses are propagated, attenuated, and structured within field-grade multi-core distribution cables. The rigorous analysis of multi-stage passive optical splitters and mathematical power link budgeting demonstrated how a network ensures operational durability while supporting thousands of subscribers.

Through practical field training under Bfone Fibernet, standard industry protocols—including precision fusion splicing, standard color coding sequence alignment, and advanced OTDR-based diagnostic fault-finding—were thoroughly mastered. Furthermore, the field study at the BSNL Exchange Office in Manjeri provided vital structural insight into macro-scale network topology, illustrating the exact handshake between hardware nodes like OLTs, BNG BSNL Exchange Office in Manjeri provided vital structural insight into macro-scale network topology, illustrating the exact handshake between hardware nodes like OLTs, BNG gateways, and consumer-end ONTs. Ultimately, this module confirmed that efficient FTTH architectures require a precise balance between strict engineering by mathematics and rigorous field deployment standards to maintain continuous, high-performance gigabit connectivity.

Part II

Electrical Home Appliances Maintenance and Troubleshooting

Part Introduction: Domestic Electrical Engineering

The integration of specialized electro-mechanical hardware remains a vital pillar of the modern domestic environment. From electrothermal conversion units to complex induction power inverters and single-phase electromagnetic distribution motors, these household systems represent direct physical applications of fundamental thermodynamic and electromagnetic theories. This second module of the report provides a rigorous technical breakdown of consumer appliance electronics, diagnostic logic, and preventive maintenance protocols executed during the industrial training.

The objective of this module is to move beyond simple component replacement to establish a thorough engineering understanding of circuit safety and appliance operations. The chapters ahead map out essential circuit structures (series vs. parallel wiring configurations) and structural protection systems—such as chassis ground earthing, Miniature Circuit Breakers (MCBs), and high-sensitivity Residual Current Circuit Breakers (RCCBs)—designed to mitigate short circuits and hazardous current leakage. Following this safety framework, the module provides an isolated component breakdown, operational analysis, and troubleshooting matrix for seven core appliances: the electric iron, induction cooker, mixer grinder, storage water geyser, ceiling fan, wall fan, and centrifugal water pump. This foundational mechanical study finishes with applied practical audits covering consumer load accounting and electrical charge tracking.

Chapter 8

Fundamentals of Electrical Systems & Safety

8.1 Introduction

The modern domestic environment relies entirely on the seamless operation of electrical appliances. From food preparation to climate control and water management, these devices have transitioned from luxuries to everyday essentials. However, continuous operation inevitably leads to mechanical wear, insulation degradation, and electrical faults.

This internship was undertaken to bridge the gap between abstract electrical theory and hands-on diagnostic engineering. The training structured under this module focused on developing a core competency in domestic wiring systems, understanding electrical safety protocols, and mastering the systematic teardown, repair, and preventive maintenance of seven foundational household appliances. The primary objective was not merely to fix broken devices, but to understand the physics governing their operation—such as electrothermal conversion, electromagnetism, and fluid dynamics—while strictly adhering to industrial safety standards.

8.2 Fundamentals of Electrical Systems & Safety Overview

Before working on any appliance, a solid foundation in circuit behavior and safety infrastructure was established.

8.2.1 Circuit Behaviors & Laws

All appliance diagnostics rely on **Ohm's Law** ($V = IR$) and the **Electrical Power Equation** ($P = VI = I^2R$). Understanding these allowed for calculating expected resistances during appliance testing to identify short or open circuits.

- **Series Connections:** Current remains constant; voltages drop across components. Used in safety switches and fuses.
- **Parallel Connections:** Voltage remains constant; current divides. Used in domestic house wiring so that the failure of one appliance does not interrupt power to the others.

8.2.2 Earthing and Protection Systems

The internship emphasized that safety systems are non-negotiable when dealing with line voltages (230V AC).

- **Earthing (Grounding):** Connects the metallic chassis of an appliance directly to the earth. In case of an insulation failure where a live wire touches the body, the current safely diverts to the ground instead of passing through a human body.
- **Fuse / MCB (Miniature Circuit Breaker):** Protects the circuit from **overcurrent** and **short circuits** by breaking the circuit when current exceeds safe thresholds.
- **ELCB / RCCB (Residual Current Circuit Breaker):** Detects minute current leakages (as low as 30mA) escaping to the ground. It compares the incoming current in the live wire against the outgoing current in the neutral wire and trips instantly if an imbalance is detected.

Chapter 9

Technical Breakdown & Diagnostics of Core Home Appliances

9.1 Electric Iron

A. Working Principle

An electric iron works on the principle of the **Joule's Heating Effect of Electric Current**. When an electric current passes through a high-resistance conductor (heating element), it generates thermal energy ($H = I^2Rt$). Temperature regulation is achieved via a **bimetric strip thermostat**. The strip consists of two different metals bonded together with distinct coefficients of thermal expansion. As the iron heats up, one metal expands faster than the other, causing the strip to bend and open the electrical contacts, breaking the circuit. Once it cools down, it straightens, closes the contact, and heat generation resumes.

B. Key Components and Parts

- **Soleplate:** The heavy, polished aluminum or cast-iron base that transfers heat directly to the clothes.
- **Heating Element:** A ribbon of **Nichrome** wound around a mica sheet insulated to prevent electric shock while maximizing heat transfer.
- **Bimetallic Thermostat:** The temperature control mechanism equipped with an adjustable knob to set specific heat thresholds.
- **Thermal Fuse:** A backup safety device that permanently melts and opens the circuit if the thermostat fails.

C. Appliance Conclusion

Through the study of the electric iron, the practical application of electrothermal conversion and thermal feedback loops via bimetallic components was fully understood. Testing this device highlighted how crucial thermal fuses are in preventing catastrophic overheating.

9.2 Induction Cooker

A. Working Principle

Unlike traditional cooking methods that use thermal conduction, an induction cooker operates on the principles of **Electromagnetism and Eddy Currents**. The appliance takes 50Hz AC power and converts it into a high-frequency (20kHz to 40kHz) alternating current using an internal inverter circuit. This high-frequency current passes through a copper induction coil beneath the ceramic plate, creating a dynamic, rapidly alternating magnetic field. When a ferromagnetic vessel is placed on top, this magnetic field induces circulating **eddy currents** within the base of the pan. Because the pan has electrical resistance, these currents generate heat directly inside the cookware itself, keeping the stovetop relatively cool.

B. Key Components and Parts

- **Induction Coil:** A flat, spirally wound copper coil that generates the alternating magnetic field.
- **IGBT (Insulated Gate Bipolar Transistor):** The high-speed electronic switch that modulates power and creates the high-frequency current.
- **Main PCB & Microcontroller:** The "brain" that monitors user inputs, manages power delivery, and reads sensor data.
- **Cooling Fan & RTD (Resistance Temperature Detector):** Critical elements for heat dissipation and monitoring pan temperatures to prevent dry boiling.

C. Appliance Conclusion

The induction cooker diagnostics provided essential insights into solid-state electronics and high-frequency power switching. It demonstrated how magnetic fields can be manipulated for localized heating, emphasizing the need for robust PCB-level troubleshooting.

9.3 Mixer Grinder

A. Working Principle

The mixer grinder operates on electromagnetic mechanical conversion using an **AC Universal Motor**. A universal motor is a series-wound motor designed to run on either AC or DC power. When voltage is applied, current flows through both the stator (field windings) and the rotor (armature windings) via carbon brushes. This creates opposing magnetic fields, producing a high starting torque and high rotational speeds (up to 20,000 RPM), which are necessary for chopping, blending, and grinding food materials.

B. Key Components and Parts

- **Stator & Rotor (Armature):** The fixed and rotating electromagnetic components that generate mechanical torque.

- **Commutator & Carbon Brushes:** The assembly that transfers electrical current from the stationary wires to the rotating armature.

9.3.1 Ceiling Fan

A. Working Principle

A ceiling fan operates on the principle of a **Single-Phase Induction Motor (Permanent Split Capacitor Motor)**. Single-phase AC power cannot generate a rotating magnetic field on its own to start a motor. To solve this, the motor has two separate windings: a **Starting Winding** and a **Running Winding**. A capacitor is wired in series with the starting winding to create a 90° electrical phase shift in the current. This phase shift creates a simulated rotating magnetic field, which induces a current in the squirrel-cage rotor, generating the torque needed to turn the fan blades.

B. Key Components and Parts

- **Stator & Rotor:** Stationary inner core with starting/running windings and the rotating outer casing.
- **Run/Start Capacitor:** Typically 2.25 μ F to 2.5 μ F capacitor used to split the phase and start rotation.
- **Ball Bearings:** Precision bearings that reduce mechanical friction, ensuring smooth and quiet operation.
- **Down-Rod Assembly:** Structural hardware used to safely suspend the fan from the ceiling.

C. Appliance Conclusion

Working with ceiling fans clarified the theory of phase splitting in single-phase induction systems. The practical experience emphasized the diagnostic process of isolating electrical winding failures from mechanical bearing resistance.

9.3.2 Wall Fan

A. Working Principle

A wall fan operates similarly to a ceiling fan, using a **Single-Phase Permanent Split Capacitor Induction Motor**, but includes specialized mechanisms for targeted directional airflow and oscillation. While the motor creates mechanical rotation using a starting winding, running winding, and a capacitor to shift the current phase, it also drives a mechanical **synchronous link gear system** or an independent **secondary micro-motor** to achieve horizontal oscillation (swiveling back and forth). Speed regulation is managed either via a multi-tapped transformer winding or an electronic TRIAC circuit on a PCB, controlling voltage delivery to the motor.

B. Key Components and Parts

- **Fan Blades and Guard:** Aerodynamically pitched plastic or metal blades protected by a wire mesh safety cage.
- **Oscillation Gear Assembly / Clutch Knob:** A mechanical gear train connected to the rotor shaft that translates high-speed rotation into slow, wide-angle swiveling motion.
- **Speed Control Switch (Pull-Cord / Push-Button / Remote PCB):** Mechanisms to toggle between different motor speed taps or control electronic voltage regulation.
- **Thermal Cut-Off (TCO):** A compact thermal fuse embedded directly inside the motor windings to protect the motor from burning out if the blades get obstructed or jammed.

C. Appliance Conclusion

The analysis of the wall fan highlighted how structural mechanisms (linkage gears) integrate with electrical induction motors. It offered an excellent case study on thermal protection embedded within motor windings and the diagnostic tracking of mechanical oscillation failures.

Chapter 10

Troubleshooting & Maintenance Practice Matrix

Appliance	Common Symptom	Root Cause Detected	Corrective Action Performed
Electric Iron	No heating; indicator light off	Open circuit in thermal fuse / Broken power cord	Replaced thermal fuse; trimmed and re-terminated power cord.
Induction Cooker	Dead display; won't power up	Blown main fuse due to shorted IGBT	Replaced the shorted IGBT and main glass fuse; checked gate driver circuit.
Mixer Grinder	Sparks from bottom; sudden stop	Worn carbon brushes / Overload switch tripped	Replaced carbon brushes; cleared jar jam and reset the bottom OLP switch.
Water Heater	Water not heating; indicator on	Scale buildup or open-circuit in heating element	Drained tank, removed limescale, and replaced the immersion heating element.
Ceiling Fan	Humming sound; spins very slowly	Degraded or leaked capacitor	Desoldered/disconnected old capacitor, verified value, and installed new $2.5\mu\text{F}$ unit.
Wall Fan	Fan rotates but does not oscillate / Overheating	Stripped linkage gears / Dry rotor shaft	Replaced the plastic gear wheel assembly; cleaned and lubricated the rotor shaft with machine oil.
Pump Motor	Motor hums but does not rotate	Seized bearings / Defective starter capacitor	Freed impeller manually; replaced bearings and capacitor.

10.1 Internship Outcomes & Skills Acquired

- **Safety Instinct Diagnostics:** Developed a strict protocol-first mindset—always verifying isolation with a neon tester, utilizing insulation mats, and ensuring appropriate grounding before applying mains voltage.
- **Systematic Fault Profiling:** Shifted away from random guesswork to structured troubleshooting: tracing faults from the source point(plug/cord) through protection system

(OLPs/fuses) to control system and finally the load.

- **Instrument Competency:** Attained fluid precision in utilizing digital multimeters to measure resistance drops, AC loop voltages, and verifying open/short circuits.

Chapter 11

Applied Practical Calculations

11.1 Domestic Electricity Consumption Auditing

To calculate household power consumption, energy is tracked in kilowatt-hours (kWh), where 1 Unit = 1 kWh.

Audit Problem:

Determine the exact 30-day billable consumption for a household running two 40W tube lights for 6 hours daily, three 65W fans for 8 hours daily, and separate appliances drawing 150W for 2 hours daily.

$$E_{\text{Lights}} = 2 \times 40 \text{ W} \times 6 \text{ hrs/day} \times 30 \text{ days} = 14,400 \text{ Wh} = 14.4 \text{ kWh}$$

$$E_{\text{Fans}} = 3 \times 65 \text{ W} \times 8 \text{ hrs/day} \times 30 \text{ days} = 46,800 \text{ Wh} = 46.8 \text{ kWh}$$

$$E_{\text{Misc}} = 150 \text{ W} \times 2 \text{ hrs/day} \times 30 \text{ days} = 9,000 \text{ Wh} = 9.0 \text{ kWh}$$

$$E_{\text{Total}} = 14.4 \text{ kWh} + 46.8 \text{ kWh} + 9.0 \text{ kWh} = \mathbf{70.2 \text{ kWh (Units)}}$$

11.2 Charge Calculation in Audio Systems

Problem:

A 230V system consumes a total of 8 MJ of energy during operation. Determine the absolute electrical charge that moved through the circuit.

Given:

$$V = 230 \text{ V}, \quad E = 8 \times 10^6 \text{ J}$$

Using the electrical work potential equation ($E = qV$):

$$q = \frac{E}{V} = \frac{8 \times 10^6 \text{ J}}{230 \text{ V}} = \mathbf{3.48 \times 10^4 \text{ Coulombs (C)}} \quad (11.1)$$

Part Conclusion: Domestic Electrical Engineering

In conclusion, this second module has established a robust technical bridge between undergraduate physics and field-level electrical repair logic. By systematically mapping out internal component frameworks across seven fundamental consumer appliances, the training provided deep insights into everyday electrical systems. Tracing diagnostic signatures—such as a bimetallic contact opening in a thermal loop or a split-phase capacitor drop in an induction motor—showed how equipment behaviors can be accurately tracked using standard multi-meter analysis.

The successful implementation of practical testing logic highlighted how embedded safety devices like thermal cut-offs (TCOs), overload protectors (OLPs), and dual-stage pressure relief valves are critical for preventing motor burnouts and protecting users. Additionally, the quantitative practical math problems confirmed that accurate load auditing and tracking energy changes are key to managing household electricity efficiently. Ultimately, the mechanical repair skills and systemic troubleshooting methods gained during this maintenance module form a critical practical toolkit, ensuring that home environments remain completely safe and operationally productive.